

UNIT 7

EARTH'S MINERALS

Learning objectives:

1. Identify minerals, rocks, and their properties.
2. Explain weathering, erosion, and rock changes.
3. Recognize everyday uses of minerals.
4. Compare soil types and their components.
5. Describe how animals and plants impact soil.
6. Analyze environmental effects on rocks and soil.
7. Apply knowledge through hands-on activities.

THE BIG IDEA: HOW CAN WE DESCRIBE ROCKS AND SOIL?



7.1 Rocks

How can we describe rocks?

A **geologist** is a scientist who studies rocks. One thing geologists look at is a rock's colour. Many rocks are one colour. Others are more than one colour. Most rocks are grey. But some rocks are black, brown, red, white, or pink.



Geologists look at the size of rocks, too. Rocks that are the same size might not weigh the same.



These two rocks are the same size.
← But the rock on the right weighs less than the rock on the left.



What are rocks made of?

All rocks are made of **minerals**. Some rocks are made of one mineral. Others are made of many minerals. Look at the piece of granite in the diagram below. It is made of three minerals. The grey parts are the mineral quartz. The white parts are feldspar. The black parts are mica.

Did you know that you use minerals every day?

A pencil lead is made of the mineral graphite. Our bodies need minerals, too. We get minerals from the foods we eat. Many toothpastes have fluoride, which is made from the mineral, fluorite.



↑ Granite



↑ Fluorite

✓ Quick Check

Fill in the blanks.

1. Granite is made of three _____.
2. The mineral in a pencil is _____.



How can we describe minerals?

A **property** tells you something about an object. Colour is one property of a mineral. Luster is another property.

Lustre tells how a mineral looks when light shines on it.

·Some minerals, like quartz, shine like glass.



quartz

·Pyrite's lustre and colour make it look like gold.



pyrite

·The mineral halloysite does not shine. It has a dull lustre.



halloysite



Another property of a mineral is its hardness. **Hardness** is how tough a mineral is. Talc is a soft mineral. You can scratch it with your fingernail. Diamond is a hard mineral. It can only be cut by another diamond.



↑ Talc is the softest mineral



↑ Diamond is the hardest mineral

Quick Check

List three properties of minerals.

Properties of Minerals.



7.2 Rocks Change

How do rocks change?

Most rocks are very hard, but they can change their size and shape. **Weathering** is the way water and wind change rocks. When water gets into the cracks of rocks, it can freeze and push against the rocks. Then the cracks get bigger and the rocks break.

Quick Check

Write true if the sentence is true. Write false if the sentence is false.

1. *Rocks can change their size.* _____
2. *Water can make rocks break.* _____
3. *When rocks slide down a hill, they get larger.* _____



↑ Strong winds can blow sand against rocks. Wind and sand made a hole in this rock.



↑ When rocks slide down a hill, they may break and become smaller.



What other ways can rocks change?

Earthquakes can change rocks, too. In an earthquake, rocks rub against each other and break into smaller pieces. Plants can also change rocks. Plants can grow in soil inside the cracks of rocks. Sometimes a plant's roots are so strong they cause the rocks to break.



↑ The roots of this tree have grown into the rock and cracked it.

You know that rocks are made of minerals. Water can cause some minerals to change.



↑ Water caused the copper in this rock to turn green.



↑ Water caused the iron in this rock to rust and turn red and brown.



7.2 Soil

What is soil made up of?

Soil is made up of tiny rocks and bits of plants and animals. Weathering makes large rocks break down into smaller rocks. They become part of the soil. When plants and animals die, they also break down and become part of the soil.



Most plants grow in soil. Plants grow by taking in minerals from the soil. People need minerals to grow, too. One way we get minerals is by eating plants.



↑ Plants get minerals from the soil.
Then people get minerals from
eating plants.



Quick Check

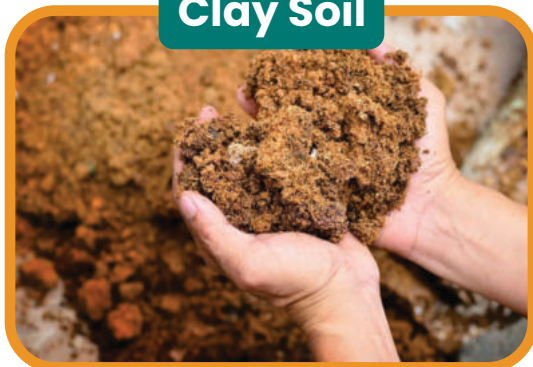
1. Name three things that are part of soil.

2. How can people get the minerals they need?

What are some kinds of soil?

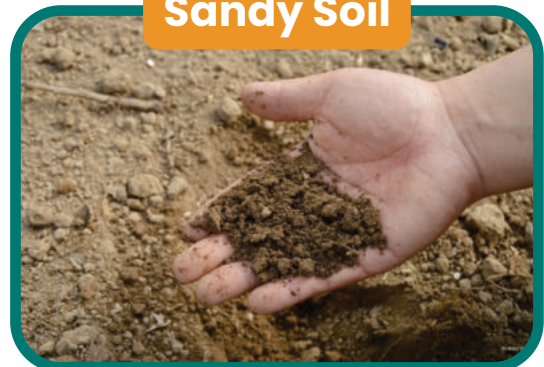
Did you know there are different kinds of soil? The minerals in the rocks give these soils their colour.

Clay Soil



↑ Red clay soil gets its color from tiny pieces of iron. It does not hold much water.

Sandy Soil



↑ Sandy soil is light brown and does not hold much water.

Most plants grow best in topsoil. Plants that do not need much water grow best in sandy soil. Some plants grow best in a mix of topsoil and sandy soil.

Topsoil

Topsoil is dark brown or black and can hold lots of water. It has bits of dead animals and plants in it.





Quick Check

Circle the answer.

1. The color of red soil comes from _____.

a iron

b plants

c animals

2. Most plants grow best in _____.

a sandy soil

b clay soil

c topsoil

How do animals help the soil?

Ants, worms, rabbits, and gophers live underground. They dig tunnels in the soil. The digging mixes the soil. This helps air and water get into the soil. The air and water help plants grow.



When animals die, their bodies break down and become part of the soil. This makes the soil healthy and helps new plants to grow.



↑ Gophers dig tunnels and mix the soil.

Quick Check

1. List three types of soil.

2. Write about two ways animals help soil.



Complete each sentence with words from the box.

geologist

hardness

lustre

property

soil

weathering

minerals

1. All rocks are made of _____.
2. Rocks are changed by _____.
3. A scientist who studies rocks is a _____.
4. Colour is an example of a _____.
5. Two properties of minerals are _____
and _____.
6. Tiny rocks are found in _____.



Test Your Understanding

Q1.What is the role of geologists in studying rocks?

Ans: _____

Q2.How do minerals affect the properties of rocks?

Ans: _____

Q3.Describe the process of weathering and its impact on rocks.

Ans: _____

Q4.Why do some minerals shine while others appear dull?

Ans: _____

Q5.How does soil help plants grow?

Ans: _____

Q6.What happens to rocks during an earthquake?

Ans: _____



Fill in the blanks.

1. Rocks are made of _____.
2. The way water and wind change rocks is called _____.
3. _____ is a scientist who studies rocks.
4. _____ is the measure of how tough a mineral is.
5. The red colour in clay soil comes from tiny pieces of _____.
6. Plants get minerals from _____ and people get minerals from _____.



Tick (✓) for True and cross (x) for False statements.

1. Soil is made up only of tiny rocks. ()

2. Lustre describes how a mineral looks when light ()

shines on it.

3. Hard minerals can be scratched easily. ()

4. Water can change minerals inside rocks. ()

5. Weathering makes rocks larger over time. ()

6. Most plants grow best in sandy soil. ()



Multiple Choice Question.

1. What property of a mineral describes how it reflects light?

- a) Hardness
- b) Lustre
- c) Weathering
- d) Soil

2. Which of the following is NOT a type of soil?

- a) Sandy soil
- b) Clay soil
- c) Iron soil
- d) Topsoil

3. What process causes rocks to break apart over time?

- a) Weathering
- b) Hardness
- c) Mining
- d) Shining

4. Which mineral property determines if a rock can be scratched?

- a) Lustre
- b) Hardness
- c) Size
- d) Weight



5. What do plants take from the soil to grow?

- a) Sand
- b) Minerals
- c) Iron
- d) Water only

6. Which soil type holds the most water?

- a) Clay soil
- b) Sandy soil
- c) Topsoil
- d) None of these





Activity: Rock Observation Experiment



ACTIVITY

- Collect three different types of rocks (smooth, rough, and shiny).
- Observe their colours, texture, and weight.
- Use a nail to gently scratch each rock and record which one is hardest.
- Compare results and discuss why some rocks are harder than others.

Soil Absorption Test

- Gather three different soil samples (sandy, clay, and topsoil).
- Pour equal amounts of water on each soil type and observe how much water they absorb.
- Discuss which soil retains the most water and why that is important for plants.

